

Homework 1 – Workflow Test

TEST ONLY – NOT FOR SUBMISSION. This short example checks the template, equations, PDF compilation, and GitHub transfer. The final solution will be written from the student's handwritten reasoning and steps.

Question 1. Two routes to the same final pressure (sample only). One mole of ideal oxygen starts at $T_1 = 300$ K and $p_1 = 1$ bar. The specified final pressure is $p_2 = 8$ bar. Take $\gamma = 1.40$ and assume a calorically perfect gas.

Sample calculation: the adiabatic work relation. For a closed system with negligible changes in bulk kinetic and potential energy, heat into the gas is positive and work done by the gas is positive. The first law is

$$dE = \delta Q - \delta W_{\text{by}} \implies \Delta E = Q - W_{\text{by}}.$$

Source: Lecture 1 organized notes, PDF page 13 (printed page 16), Eqs. (21)–(24). The finite-process form follows by integration. Here E denotes the total internal energy of the gas.

For an adiabatic process, $Q = 0$. Rearranging the first law therefore gives

$$\Delta E = -W_{\text{by,ad}} \implies W_{\text{by,ad}} = -\Delta E.$$

For a calorically perfect gas, subtract the endpoint values in $e = c_v T + e_{\text{ref}}$ and multiply by the fixed mass m :

$$\Delta E = mc_v(T_2 - T_1) = n\bar{c}_v(T_2 - T_1).$$

Source: Lecture 3 organized notes, PDF page 1 (printed page 25), Eq. (74), followed by endpoint subtraction and a basis conversion. Here e and c_v are mass-specific, n is the amount in moles, and $\bar{c}_v$ is the molar heat capacity; by definition $mc_v = n\bar{c}_v$.

Substituting this energy change into the adiabatic first law gives

$$W_{\text{by,ad}} = -n\bar{c}_v(T_2 - T_1).$$

This is a combination of the two cited relations. It is not a separate formula quoted directly from the notes. If compression raises the temperature, work done by the gas is negative.

Status: This is an incomplete test excerpt, not an answer to all parts of Problem 1. Test text and LaTeX preparation were assisted by ChatGPT. The final submission will include an accurate acknowledgment of the assistance actually used.